



(10) Publication No: **IN202641027783 A1**

(22) Date of Filing: 09-03-2026

(43) Publication Date: 20-03-2026

Journal No: 12/2026

(19) Intellectual Property Office, India

(12) INDIAN PATENT APPLICATION

(54) Title: **Multi-domain feature fusion based deep learning system for audio deepfake detection**

(51) International Classification:

**G10L 25/30, G06N 3/04, G06N 3/08, G10L 15/16
G10L 25/51**

(21) Application No: **202641027783**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --

Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --

Filing Date: --

(62) Divisional to Application No: --

Filing Date: --

(71) Name of Applicant(s):

Vellore Institute Of Technology,

(72) Name of Inventor(s):

1. Dr. Mohanasundaram R

2. Rayban Pranav Mahesh

3. Ajitesh Sharma

4. Hansaj Patidar

(57) Abstract:

ABSTRACT MULTI-DOMAIN FEATURE FUSION BASED DEEP LEARNING SYSTEM FOR AUDIO DEEPFAKE DETECTION The present disclosure provides a computer-implemented method (100) for audio deepfake detection that includes receiving (102) an audio input, extracting (106) spectral features from a spectral representation of the audio input using a convolutional neural network (64108), extracting (108) acoustic features from cepstral coefficients of the audio input using a recurrent neural network (64110), extracting (110) temporal features from a raw waveform of the audio input using a one-dimensional convolutional neural network (64112), fusing (112) the spectral features, the acoustic features, and the temporal features using a self-attention module (64114) to generate a fused feature representation, classifying (114) the fused feature representation using a neural network classifier (64116), and outputting (116) a classification indicating whether the audio input is real or deepfake along with a confidence score. (FIG. 1) [To be published along with figure 1]

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

5

&

THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

10

(See section 10 and rule 13)

TITLE OF INVENTION

MULTI-DOMAIN FEATURE FUSION BASED DEEP LEARNING SYSTEM
FOR AUDIO DEEPPFAKE DETECTION

15

APPLICANT NAME AND ADDRESS

Vellore Institute of Technology

20

An Indian University having address as:

Vellore Campus, Vellore - 632 014

Tamil Nadu, India

THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES

25

THE INVENTION AND THE MANNER IN WHICH IT IS TO BE

PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to artificial intelligence, machine learning, digital signal processing, audio forensics, and cybersecurity, and more
10 particularly to a multi-domain feature fusion based deep learning system for robust audio deepfake detection that integrates time-domain, frequency-domain, cepstral, and statistical acoustic features with adaptive attention-based fusion for discriminating between real and synthetic audio signals.

BACKGROUND

15 [0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Audio deepfake detection has emerged as a field of increasing relevance with the proliferation of synthetic speech generation technologies. Conventional
20 approaches to detecting synthetic or manipulated audio have employed various signal processing and machine learning techniques. Research in this area has explored CNN-based spectrogram analysis for automatic speaker verification spoofing countermeasures, though such approaches have relied primarily on spectral features with limited generalization to unseen deepfake attacks. Other work
25 has investigated the use of Mel-frequency cepstral coefficients (MFCC) combined with Gaussian mixture models for detecting synthetic speech, employing hand-crafted acoustic features that have demonstrated reduced robustness to noise and modern neural text-to-speech systems. End-to-end neural audio spoofing detection using raw waveform processing through one-dimensional CNNs has been explored,
30 though such approaches have exhibited high computational cost and limited multi-domain interpretability. Transformer-based audio deepfake detection methods

utilizing attention mechanisms have been developed, though these have required large datasets and have shown sensitivity to domain shift. Linear frequency cepstral coefficient approaches for generalized audio deepfake detection have demonstrated performance degradation under real-world noise and compression conditions.

5 Patent literature has disclosed systems for detecting synthetic audio signals using single-domain spectral anomaly detection (US Patent: System and Method for Detecting Synthetic Audio Signals), audio authentication using machine learning-based fingerprinting (WO Patent: Audio Authentication Using Machine Learning), and speech spoofing detection using cepstral features with classifiers (IN Patent: 10 Method for Speech Spoofing Detection). Additionally, IN-202321088542-A has disclosed a system utilizing WaveNet-based feature extraction with hybrid CNN-LSTM networks and multi-head self-attention for audio deepfake detection, while CN-120783799-A has disclosed an audio deepfake detection method integrating multi-source features with cross-scale modeling using channel attention and gating 15 mechanisms.

[0006] Existing audio deepfake detection systems face several technical challenges. Systems relying on single-domain feature representations, such as spectrograms alone or cepstral coefficients alone, have exhibited degraded performance when encountering diverse attack types or real-world acoustic 20 conditions. Simple feature concatenation approaches without adaptive weighting or attention mechanisms have failed to leverage the complementary discriminative information available across different acoustic domains. Furthermore, systems lacking frame-level temporal analysis capabilities have been unable to detect localized anomalies within longer audio streams, and approaches dependent on data 25 augmentation for robustness have shown performance that varies based on training noise characteristics rather than architectural robustness.

[0007] It has been appreciated that a system and method is needed that overcomes one or more of these problems.

[0008] The present invention addresses the above shortcomings of the prior art. 30 However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0009] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0010] In a first aspect, a method for audio deepfake detection is provided. The method comprises: receiving raw audio input; preprocessing the audio input through noise handling, normalization, and segmentation; extracting a Mel-spectrogram from the preprocessed audio and processing the Mel-spectrogram via a convolutional neural network to generate spectral features; extracting Mel-frequency cepstral coefficients with delta and delta-delta derivatives from the preprocessed audio and processing the Mel-frequency cepstral coefficients via a bidirectional long short-term memory network to generate acoustic features; processing a raw waveform of the preprocessed audio through a one-dimensional convolutional neural network to generate temporal features; fusing the spectral features, the acoustic features, and the temporal features via a multi-head self-attention module to generate a fused feature representation; classifying the fused feature representation via a deep neural network with regularization; and outputting a classification indicating whether the audio input is real or deepfake along with an associated confidence score.

[0011] The extraction and processing of features from multiple acoustic domains simultaneously, including spectral, acoustic, and temporal domains, enables the capture of complementary discriminative information that would not be available from any single feature representation alone. The use of a multi-head self-attention module for feature fusion allows adaptive weighting of features from different domains based on audio characteristics, thereby improving detection accuracy across diverse attack types and acoustic conditions compared to simple concatenation approaches.

- [0012] The method may further comprise generating the Mel-spectrogram with 128 mel bands.
- [0013] The use of 128 mel bands provides sufficient frequency resolution to capture fine-grained spectral patterns characteristic of synthetic audio generation artifacts while maintaining computational efficiency suitable for real-time processing applications.
- [0014] The method may further comprise computing the Mel-frequency cepstral coefficients with 40 coefficients.
- [0015] Computing 40 Mel-frequency cepstral coefficients captures detailed acoustic characteristics of the audio signal, enabling the bidirectional long short-term memory network to learn temporal dynamics and patterns that distinguish authentic speech from synthetically generated audio.
- [0016] The method may further comprise sampling the raw waveform with 64,000 samples for processing through the one-dimensional convolutional neural network.
- [0017] Sampling 64,000 samples from the raw waveform provides sufficient temporal resolution to capture fine-grained artifacts introduced by neural audio synthesis techniques that may not be visible in frequency-domain representations, thereby enhancing detection capability for modern deepfake generation methods.
- [0018] The method may further comprise assigning domain-aware weights to the spectral features, the acoustic features, and the temporal features, wherein the spectral features are assigned a weight of 42 percent, the acoustic features are assigned a weight of 35 percent, and the temporal features are assigned a weight of 23 percent.
- [0019] The domain-aware weighting mechanism prioritizes feature contributions based on their empirically determined discriminative power for deepfake detection, with spectral features providing the highest contribution followed by acoustic and temporal features, thereby optimizing the fusion process for maximum detection accuracy.
- [0020] The method may further comprise segmenting the audio input into temporal frames, generating a deepfake probability score for each frame, applying

a threshold of 0.5 to the frame predictions, and aggregating the frame scores using at least one of mean, median, max, or voting methods to compute an overall confidence score.

5 **[0021]** Frame-level temporal analysis enables detection of localized anomalies within longer audio streams that may contain partial deepfake content, while the multiple aggregation strategies provide robust decision-making that accounts for temporal variations in deepfake probability across the audio signal.

10 **[0022]** In a second aspect, an audio deepfake detection system is provided. The system comprises: an audio input module configured to receive raw audio input; a preprocessing module configured to perform noise handling, normalization, and segmentation on the raw audio input; a multi-branch feature extraction module comprising: a first branch comprising a convolutional neural network module configured to process a Mel-spectrogram representation of the audio to generate spectral features, a second branch comprising a bidirectional long short-term
15 memory module configured to process Mel-frequency cepstral coefficients with delta derivatives to generate acoustic features, and a third branch comprising a one-dimensional convolutional neural network module configured to process raw waveform data to generate temporal features; a multi-head self-attention fusion module configured to combine the spectral features, the acoustic features, and the
20 temporal features using an attention-based mechanism that enables adaptive weighting of features from different domains; a classification module comprising a deep neural network with regularization configured to classify the combined features as real or deepfake; and an output module configured to generate a classification result along with an associated confidence score.

25 **[0023]** The parallel multi-branch architecture enables simultaneous extraction of features from spectral, acoustic, and temporal domains, reducing processing latency compared to sequential approaches while ensuring comprehensive analysis of different audio characteristics. The attention-based fusion mechanism dynamically adjusts feature contributions based on the specific characteristics of
30 each audio sample, providing adaptability to varying audio quality and attack types without requiring retraining.

[0024] The system may further comprise a Mel-spectrogram generator configured to generate the Mel-spectrogram representation with 128 mel bands.

[0025] The dedicated Mel-spectrogram generator with 128 mel bands ensures consistent spectro-temporal feature extraction that captures frequency patterns indicative of synthetic audio generation, providing a standardized input representation for the convolutional neural network module.

[0026] The system may further comprise an MFCC extractor configured to compute the Mel-frequency cepstral coefficients with 40 coefficients along with delta and delta-delta derivatives.

[0027] The MFCC extractor with 40 coefficients and derivative calculations captures both static and dynamic acoustic characteristics, enabling the bidirectional long short-term memory module to model temporal dependencies that reveal inconsistencies in synthetic speech production.

[0028] The system may further comprise a raw waveform sampler configured to sample 64,000 samples from the raw audio input for processing by the one-dimensional convolutional neural network module.

[0029] The raw waveform sampler provides direct access to time-domain signal characteristics without transformation losses, allowing the one-dimensional convolutional neural network to detect subtle waveform artifacts that may be obscured in frequency-domain representations.

[0030] The system may further comprise a statistical descriptor module configured to aggregate multi-domain representations and compute statistical descriptors from the combined features.

[0031] The statistical descriptor module provides additional discriminative information by computing aggregate statistics across the multi-domain features, enhancing the robustness of the classification decision by capturing global characteristics of the audio signal.

[0032] In a third aspect, a non-transitory computer-readable medium storing instructions is provided. The instructions, when executed by a processor, cause the processor to: receive raw audio input; preprocess the audio input through noise handling, normalization, and segmentation; extract a Mel-spectrogram from the

preprocessed audio and process the Mel-spectrogram via a convolutional neural network to generate spectral features; extract Mel-frequency cepstral coefficients with delta and delta-delta derivatives from the preprocessed audio and process the Mel-frequency cepstral coefficients via a bidirectional long short-term memory
5 network to generate acoustic features; process a raw waveform of the preprocessed audio through a one-dimensional convolutional neural network to generate temporal features; fuse the spectral features, the acoustic features, and the temporal features via a multi-head self-attention module to generate a fused feature representation; classify the fused feature representation via a deep neural network
10 with regularization; and output a classification indicating whether the audio input is real or deepfake along with an associated confidence score.

[0033] The computer-readable medium enables deployment of the multi-domain feature fusion deepfake detection methodology across diverse computing platforms, facilitating integration into digital forensics platforms, media
15 verification systems, and cybersecurity applications while maintaining consistent detection performance.

[0034] The instructions may further cause the processor to assign domain-aware weights to the spectral features, the acoustic features, and the temporal features based on their relative discriminative importance for deepfake detection.

20 **[0035]** Domain-aware weight assignment based on discriminative importance ensures that the fusion process optimally combines features according to their contribution to detection accuracy, improving overall system performance without requiring manual tuning for different deployment scenarios.

[0036] The instructions may further cause the processor to segment the audio
25 input into temporal frames, generate a deepfake probability score for each frame, and aggregate the frame scores to determine a final classification.

[0037] Frame-level processing with score aggregation enables the detection system to identify both fully synthetic audio and partially manipulated audio streams, providing comprehensive protection against diverse audio deepfake attack
30 scenarios encountered in real-world applications.

[0038] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

5 **[0039]** The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

10 **[0040]** FIG. 1 illustrates a flowchart for a method for deepfake audio detection using multi-branch feature extraction and fusion, according to aspects of the present disclosure.

[0041] FIG. 2 illustrates a flowchart for a method for multi-domain feature extraction from a preprocessed audio signal, according to an embodiment.

15 **[0042]** FIG. 3 illustrates a flowchart for a feature fusion process with multi-head self-attention and domain-aware weighting, according to aspects of the present disclosure.

[0043] FIG. 4 illustrates a flowchart for an audio deepfake detection method with frame-level processing and aggregation, according to an embodiment.

20 **[0044]** FIG. 5 illustrates a block diagram of an audio deepfake detection system, according to aspects of the present disclosure.

[0045] FIG. 6 illustrates a block diagram of a feature extraction architecture for multi-domain audio processing, according to an embodiment.

25 **[0046]** FIG. 7 illustrates visualizations of audio waveform analysis, frame-level deepfake probability, and aggregated prediction scores, according to aspects of the present disclosure.

[0047] FIG. 8 illustrates visualizations of model accuracy versus noise level and an accuracy degradation curve, according to an embodiment.

30 **[0048]** FIG. 9 illustrates an accuracy degradation curve showing detection accuracy versus signal-to-noise ratio, according to aspects of the present disclosure.

[0049] FIG. 10 illustrates a bar chart depicting relative importance of different feature domains for deepfake detection, according to an embodiment.

[0050] FIG. 11 illustrates visualizations of feature domain importance and cross-branch attention weights, according to aspects of the present disclosure.

5 [0051] FIG. 12 illustrates training and validation performance curves for a deep learning model, according to an embodiment.

[0052] FIG. 13 illustrates a detection error tradeoff curve for the audio deepfake detection system, according to aspects of the present disclosure.

10 [0053] FIG. 14 illustrates a graphical analysis of false acceptance rate versus false rejection rate as a function of decision threshold, according to an embodiment.

[0054] FIG. 15 illustrates a precision-recall curve for the audio deepfake detection system, according to aspects of the present disclosure.

[0055] FIG. 16 illustrates a receiver operating characteristic curve for the audio deepfake detection system, according to an embodiment.

15 [0056] FIG. 17 illustrates a normalized confusion matrix for audio deepfake classification, according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0057] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

20 [0058] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

25 [0059] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit

the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0060] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0061] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures 1. In Figures 1, the same elements or elements that have the same functions are indicated by the same reference signs.

[0062] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0063] FIG. 1 illustrates a method 100 for deepfake audio detection using multi-branch feature extraction and fusion according to various embodiments. FIG. 5 illustrates an Audio Deepfake Detection System 64100 according to various embodiments.

[0064] The Audio Deepfake Detection System 64100 includes an Audio Input Module 64102, a Preprocessing Module 64104, a Multi-Branch Feature Extraction Module 64106, a Multi-Head Self-Attention Fusion Module 64114, a Classification Module 64116, and an Output Module 64118. The Audio Deepfake Detection System 64100 is configured to process audio through parallel spectral, acoustic, and temporal branches simultaneously.

[0065] The Audio Input Module 64102 is configured to receive raw audio input and forward the raw audio input to the Preprocessing Module 64104. The Preprocessing Module 64104 is configured to perform noise handling, normalization, and segmentation of the input audio signal before feature extraction.

5 The Preprocessing Module 64104 outputs preprocessed audio data to the Multi-Branch Feature Extraction Module 64106.

[0066] The Multi-Branch Feature Extraction Module 64106 includes three parallel processing branches. A Branch A - CNN Module 64108 is configured to process Mel-spectrogram representations of the audio through a convolutional
10 neural network. A Branch B - BiLSTM Module 64110 is configured to process Mel-frequency cepstral coefficients with delta derivatives through a bidirectional long short-term memory network. A Branch C - 1D-CNN Module 64112 is configured to process raw waveform data through a one-dimensional convolutional neural network. The three branches operate concurrently to extract spectral,
15 acoustic, and temporal features respectively from the preprocessed audio signal.

[0067] The outputs from the Branch A - CNN Module 64108, the Branch B - BiLSTM Module 64110, and the Branch C - 1D-CNN Module 64112 converge at the Multi-Head Self-Attention Fusion Module 64114. The Multi-Head Self-Attention Fusion Module 64114 is configured to combine heterogeneous features
20 from the three branches using an attention-based mechanism that enables adaptive weighting of features from different domains.

[0068] The fused feature representation from the Multi-Head Self-Attention Fusion Module 64114 is forwarded to the Classification Module 64116. The Classification Module 64116 is configured to employ a deep neural network with
25 regularization to classify the audio as either real or deepfake. The classification result along with an associated confidence score is transmitted to the Output Module 64118. The Output Module 64118 is configured to generate the final detection output.

[0069] The method 100 begins with a step 102, where raw audio input is
30 received. The method 100 proceeds to a step 104, where the audio is preprocessed

through noise handling, normalization, and segmentation. Following preprocessing, the method 100 branches into three parallel processing paths.

[0070] In a step 106, a Mel-spectrogram is extracted and processed via a convolutional neural network in Branch A. Concurrently, in a step 108, Mel-frequency cepstral coefficients with deltas are extracted and processed via a
5 bidirectional long short-term memory network in Branch B. Simultaneously, in a step 110, the raw waveform is processed through a one-dimensional CNN in Branch C.

[0071] The outputs from the three parallel branches converge at a step 112,
10 where the features are fused via a multi-head self-attention module. The method 100 advances to a step 114, where the fused features are classified via a deep neural network with regularization. The method 100 concludes at a step 116, where the output is generated as a Real or Deepfake classification with an associated confidence score.

[0072] The Audio Deepfake Detection System 64100 is implemented using
15 Python and PyTorch as the deep learning framework. The Audio Deepfake Detection System 64100 is optimized for reproducibility, scalability, and extensibility. The Audio Deepfake Detection System 64100 is designed for integration into digital forensics platforms, media verification systems, and
20 cybersecurity applications.

[0073] FIG. 6 illustrates a Feature Extraction Architecture 64200 for multi-domain audio processing according to various embodiments. FIG. 2 illustrates a method 200 for multi-domain feature extraction from a preprocessed audio signal according to various embodiments.

[0074] The Feature Extraction Architecture 64200 comprises a Spectral
25 Domain Processor 64202, an Acoustic Domain Processor 64208, and a Temporal Domain Processor 64214, each configured to extract distinct types of features from the audio signal. The Feature Extraction Architecture 64200 further includes a Statistical Descriptor Module 64220 and a Feature Vector Output 64222.

[0075] The Spectral Domain Processor 64202 contains a Mel-Spectrogram
30 Generator 64204 and a CNN Feature Extractor 64206. The Mel-Spectrogram

Generator 64204 is configured to convert the audio input into a Mel-spectrogram representation with 128 mel bands. The CNN Feature Extractor 64206 is configured to process the Mel-spectrogram representation to extract spectral features through convolutional neural network operations. The spectro-temporal features extracted
5 by the Spectral Domain Processor 64202 contribute the highest discriminative power among all feature domains.

[0076] The Acoustic Domain Processor 64208 contains an MFCC Extractor 64210 and a BiLSTM Processor 64212. The MFCC Extractor 64210 is configured to compute 40 Mel-frequency cepstral coefficients from the audio signal along with
10 delta and delta-delta derivatives. The BiLSTM Processor 64212 is configured to process the coefficients using bidirectional long short-term memory networks to capture acoustic feature patterns and temporal dynamics.

[0077] The Temporal Domain Processor 64214 contains a Raw Waveform Sampler 64216 and a 1D-CNN Processor 64218. The Raw Waveform Sampler
15 64216 is configured to sample 64,000 samples from the raw audio waveform. The 1D-CNN Processor 64218 is configured to process the sampled waveform through one-dimensional convolutional neural network operations to extract temporal features and capture fine-grained artifacts.

[0078] The outputs from the CNN Feature Extractor 64206, the BiLSTM
20 Processor 64212, and the 1D-CNN Processor 64218 are directed to the Statistical Descriptor Module 64220. The Statistical Descriptor Module 64220 is configured to aggregate the multi-domain representations and compute statistical descriptors from the combined features. The Statistical Descriptor Module 64220 provides the processed features to the Feature Vector Output 64222. The Feature Vector Output
25 64222 is configured to output multi-domain feature vectors for subsequent fusion and classification stages in the deepfake detection pipeline.

[0079] The method 200 begins with a step 202, where the system receives a preprocessed audio signal from the Preprocessing Module 64104. Following receipt of the preprocessed audio signal, the method 200 proceeds through parallel
30 processing branches for extracting different types of features from the audio signal.

[0080] In a step 204, the system samples a raw waveform comprising 64,000 samples for 1D-CNN processing. Concurrently, in a step 206, the system generates a Mel-spectrogram with 128 mel bands for CNN processing. In a step 208, the system computes MFCC with 40 coefficients along with delta and delta-delta derivatives.

[0081] The MFCC features computed in the step 208 are then processed via BiLSTM in a step 210. The outputs from the step 204, the step 206, and the step 210 converge at a step 212, where the system extracts statistical descriptors from the audio signal. The step 212 aggregates the multi-domain representations derived from the temporal, spectral, and cepstral processing branches.

[0082] The method 200 concludes with a step 214, where the system outputs multi-domain feature vectors for the fusion stage. The feature vectors encompass the raw waveform characteristics, Mel-spectrogram representations, MFCC-based acoustic features processed through BiLSTM, and statistical descriptors, providing a comprehensive multi-domain representation of the input audio signal for subsequent deepfake detection processing by the Multi-Head Self-Attention Fusion Module 64114.

[0083] FIG. 3 illustrates a method 300 for feature fusion with multi-head self-attention and domain-aware weighting according to various embodiments. FIG. 4 illustrates a method 400 for frame-level processing and aggregation according to various embodiments.

[0084] The method 300 begins with a step 302, where feature vectors are received from all three extraction branches of the Multi-Branch Feature Extraction Module 64106. The feature vectors include spectral features from the Branch A - CNN Module 64108, acoustic features from the Branch B - BiLSTM Module 64110, and temporal features from the Branch C - 1D-CNN Module 64112.

[0085] The method 300 proceeds to a step 304, where heterogeneous features from the different branches are concatenated. Following concatenation, the method 300 advances to a step 306, where normalization of the concatenated features is performed. The normalization prepares the concatenated features for processing by the attention mechanism.

[0086] The method 300 continues to a step 308, where multi-head self-attention is applied. The step 308 includes query, key, and value generation from the normalized concatenated features. The Multi-Head Self-Attention Fusion Module 64114 uses adaptive multi-head self-attention based feature fusion to
5 combine features from the multiple branches.

[0087] The method 300 proceeds to a step 310, where attention weights and cross-branch attention are computed based on the self-attention mechanism. The system includes a domain-aware feature weighting mechanism that dynamically prioritizes feature branches based on audio characteristics.

10 [0088] The method 300 advances to a step 312, where domain-aware weights are assigned. Spectral features receive 42 percent weighting, acoustic features receive 35 percent weighting, and temporal features receive 23 percent weighting. The domain-aware weights reflect the relative discriminative contribution of each feature domain to deepfake detection.

15 [0089] The method 300 continues to a step 314, where the weighted features are aggregated into a unified representation. The step 314 combines the attention-weighted features from all three domains into a single fused feature vector. The method 300 concludes at a step 316, where the fused feature vector is output for classification by the Classification Module 64116.

20 [0090] The method 400 begins with a step 402, where an audio stream input is received. The method 400 proceeds to a step 404, where the audio is segmented into temporal frames for individual analysis. Each temporal frame represents a discrete portion of the audio stream for frame-level processing.

[0091] The method 400 advances to a step 406, where each frame is processed
25 and a deepfake probability score is generated for that frame. The frame-level processing enables fine-grained temporal anomaly detection within audio streams.

[0092] The method 400 continues to a step 408, where a threshold of 0.5 is applied to the frame predictions to determine preliminary classification at the frame level. Frames with probability scores above the threshold are preliminarily
30 classified as deepfake, while frames with probability scores below the threshold are preliminarily classified as real.

[0093] The method 400 proceeds to a step 410, where the individual frame scores are aggregated using various methods. The system implements robust decision aggregation strategies including mean, median, max, and voting for frame-level predictions. The aggregation step combines the frame-level predictions into a unified assessment of the audio stream.

[0094] The method 400 advances to a step 412, where an overall confidence score is computed based on the aggregated results. The overall confidence score represents the system's certainty in the final classification decision.

[0095] At a step 414, the method 400 reaches a decision point where the system determines whether the aggregated score is greater than or equal to a threshold. If the aggregated score meets or exceeds the threshold, the method 400 proceeds to a step 416, where the audio is classified as DEEPFAKE along with an associated confidence score. If the aggregated score falls below the threshold, the method 400 proceeds to a step 418, where the audio is classified as REAL along with an associated confidence score.

[0096] The Classification Module 64116 includes regularization techniques to prevent overfitting during training. The regularization techniques enable the Classification Module 64116 to generalize across diverse datasets and attack types while maintaining detection accuracy under varying conditions.

[0097] FIG. 7 illustrates frame-level temporal deepfake probability analysis and decision aggregation according to various embodiments.

[0098] The top panel of FIG. 7 displays an audio waveform spanning approximately 3.5 seconds. The audio waveform exhibits amplitude variations between negative 0.4 and positive 0.4. The waveform demonstrates characteristics of a speech signal with varying intensity patterns throughout the duration of the audio sample.

[0099] The middle panel of FIG. 7 illustrates frame-level deepfake probability analysis over the same temporal range as the audio waveform. A horizontal dashed line at 0.5 marks the decision boundary threshold. The probability curve fluctuates across the timeline with most values remaining below the threshold in a region indicating authentic audio. A notable spike occurs around the 2.0-second mark

where the probability exceeds 0.8, briefly entering a region above the threshold that indicates potential deepfake content. The probability scale ranges from 0.0 to 1.0, transitioning from values indicating authentic audio through intermediate values to values indicating potential deepfake content.

5 **[0100]** The bottom panel of FIG. 7 presents aggregated prediction scores using four different aggregation methods. The mean aggregation method produces a score of 0.077. The median aggregation method produces a score of 0.085. The max aggregation method produces a score of 0.732. The min aggregation method produces a score of 0.148. A vertical dashed line marks the 0.5 decision threshold.

10 The final classification result is REAL based on the aggregated scores, as the mean, median, and min values fall below the decision threshold despite the max value exceeding the threshold. The aggregation approach enables robust classification by considering multiple statistical measures of the frame-level predictions rather than relying on a single aggregation method.

15 **[0101]** FIG. 10 illustrates the relative importance of different feature domains for deepfake detection according to various embodiments.

[0102] The chart in FIG. 10 displays three distinct feature extraction branches with their respective relative importance values. The Mel Spectrogram branch, representing spectro-temporal features, demonstrates a relative importance of 42.0

20 percent. The Mel Spectrogram branch utilizes 128 mel bands and captures frequency patterns from the audio signal. The MFCC plus Deltas branch, representing acoustic features, demonstrates a relative importance of 35.0 percent. The MFCC plus Deltas branch employs 40 coefficients plus delta and delta-delta calculations to capture temporal dynamics. The Raw Waveform branch,

25 representing time-domain features, demonstrates a relative importance of 23.0 percent. The Raw Waveform branch processes 64,000 samples and captures fine-grained artifacts from the audio signal.

[0103] The distribution of importance across all three domains demonstrates that each feature domain provides complementary discriminative information for

30 deepfake detection. The spectro-temporal features offer the greatest contribution to deepfake detection accuracy, followed by acoustic cepstral features and raw

temporal waveform analysis. The multi-branch architecture extracts and combines features from all three domains rather than relying on any single feature representation alone.

5 **[0104]** FIG. 11 illustrates gradient-based feature importance analysis and cross-branch attention weights according to various embodiments.

[0105] The left portion of FIG. 11 displays feature domain importance based on gradient-based analysis. The system uses gradient-based feature importance analysis to visualize attention weights learned by the fusion module. The Mel Spectrogram CNN branch demonstrates a relative importance of 42.00 percent. The MFCC BiLSTM branch demonstrates a relative importance of 35.00 percent. The Raw Waveform 1D CNN branch demonstrates a relative importance of 23.00 percent.

10 **[0106]** The right portion of FIG. 11 depicts cross-branch attention weights illustrating how each query branch attends to information from all three feature domains including Mel, MFCC, and Raw representations. The attention weight values demonstrate adaptive domain weighting based on audio characteristics. When Mel features serve as the query, the attention weights are 0.42 for Mel, 0.33 for MFCC, and 0.25 for Raw. When MFCC features serve as the query, the attention weights are 0.47 for Mel, 0.32 for MFCC, and 0.21 for Raw. When Raw features serve as the query, the attention weights are 0.46 for Mel, 0.36 for MFCC, and 0.19 for Raw.

15 **[0107]** The cross-branch attention weights demonstrate that the attention mechanism dynamically prioritizes different feature branches based on audio characteristics. The adaptive domain weighting mechanism forms a component of the attention-based fusion module within the deepfake detection architecture. The gradient-based feature importance analysis provides interpretability by quantifying the contribution of each feature domain to the final classification decision.

20 **[0108]** FIG. 8 illustrates adversarial robustness characteristics of the audio deepfake detection system under varying noise conditions according to various embodiments.

[0109] The left panel of FIG. 8 displays a bar chart titled "Model Accuracy vs Noise Level" showing classification accuracy percentages across different signal-to-noise ratio conditions. The bar chart presents performance measurements at five distinct noise conditions. At Clean conditions, the system achieves 100.0% accuracy. At 20 dB SNR, the system achieves 96.2% accuracy. At 10 dB SNR, the system achieves 94.4% accuracy. At 5 dB SNR, the system achieves 92.5% accuracy. At 0 dB SNR, the system maintains accuracy above the minimum threshold. The bars transition in color from green for higher accuracy values to orange and red for lower values. A horizontal dashed red line indicates the 90% target threshold, demonstrating that the system maintains accuracy above 90% for moderate noise levels.

[0110] The right panel of FIG. 8 presents an Accuracy Degradation Curve plotting accuracy percentage against SNR in decibels. The curve demonstrates how model performance degrades as noise increases from high SNR values toward 0 dB. The system maintains stable accuracy above 90% for moderate noise levels before declining more steeply at extreme noise conditions. Two horizontal reference lines are displayed on the curve: a dashed red line at 90% representing the target accuracy and a pink shaded region below 80% indicating the acceptable performance boundary. Blue circular markers along the curve indicate measured data points at specific SNR values.

[0111] FIG. 9 illustrates an Accuracy Degradation Curve demonstrating the relationship between Signal-to-Noise Ratio and detection accuracy according to various embodiments.

[0112] The horizontal axis of FIG. 9 displays SNR values ranging from 100 dB on the left to 0 dB on the right, representing progressively increasing noise conditions. The vertical axis shows accuracy percentages from 0 to 100 percent. A blue line with circular data point markers tracks the system's performance across different noise levels. Two horizontal reference lines are displayed: a red dashed line at 90 percent labeled as the Target threshold and an orange dashed line at 80 percent labeled as the Acceptable threshold.

[0113] The curve demonstrates that the system maintains near-perfect accuracy of approximately 100 percent at high SNR values from 100 dB down to approximately 20 dB. As SNR decreases below 20 dB, the accuracy begins to decline, dropping to approximately 97 percent at around 15 dB, then falling to approximately 85 percent at around 8 dB, and reaching approximately 82 percent at the lowest SNR values near 0 dB. The system achieves graceful performance decay instead of abrupt failure under extreme noise conditions, maintaining accuracy above the acceptable 80 percent threshold even under extreme noise scenarios.

10 [0114] The adversarial robustness evaluation validates that the multi-domain feature fusion system maintains robust detection performance above the 90% accuracy threshold across moderate noise conditions. The graceful degradation characteristic rather than catastrophic failure under adverse noise conditions demonstrates suitability for real-world forensic and security deployment scenarios

15 where audio quality is compromised. The robustness under varying signal-to-noise ratios enables deployment in adversarial acoustic environments where audio signals are subject to noise interference, compression artifacts, or intentional perturbations.

[0115] FIG. 12 illustrates training and validation performance metrics over epochs according to various embodiments.

20 [0116] The Audio Deepfake Detection System 64100 includes an Evaluation Module that computes Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics during training and validation phases. The training datasets include text-to-speech, voice conversion, and neural audio synthesis attack samples to ensure comprehensive coverage of diverse deepfake generation techniques.

25 [0117] FIG. 12 presents six performance metric plots arranged in a two-by-three grid, displaying the training and validation curves over approximately 50 epochs. Each plot contains two lines, with a blue line representing training performance and a red line representing validation performance.

[0118] The top row of FIG. 12 displays Loss, Accuracy, and AUC-ROC metrics. The Loss plot shows both training and validation loss decreasing rapidly from initial values around 0.4-0.5 to stabilize near 0.1, with the curves converging

30

closely after approximately 10 epochs. The Accuracy plot demonstrates both metrics rising sharply from around 50% to approximately 95-100%, with training accuracy slightly exceeding validation accuracy throughout training. The AUC-ROC plot shows rapid improvement from approximately 0.5 to values approaching 1.0, with both curves stabilizing at high values after the initial epochs.

[0119] The bottom row of FIG. 12 presents Precision, Recall, and F1-Score metrics. The Precision plot shows initial fluctuation with values starting around 30-40% before stabilizing near 95-100% for both training and validation. The Recall plot indicates that training recall reaches and maintains approximately 100%, while validation recall shows more variation but generally remains high around 90-100%. The F1-Score plot demonstrates improvement from approximately 50% to values stabilizing around 95-100%, with some oscillation in the validation curve.

[0120] The close tracking between training and validation curves demonstrates absence of overfitting and confirms effective training with strong generalization capability of the multi-domain feature fusion neural network for audio deepfake detection.

[0121] FIG. 13 illustrates a Detection Error Tradeoff curve according to various embodiments.

[0122] The Detection Error Tradeoff curve in FIG. 13 illustrates the relationship between False Acceptance Rate and False Rejection Rate for the audio deepfake detection system. The horizontal axis represents the False Acceptance Rate in percentage, displayed on a logarithmic scale ranging from 0.1 to 50 percent. The vertical axis represents the False Rejection Rate in percentage, also on a logarithmic scale spanning from 0.1 to approximately 99 percent.

[0123] The blue curve traces the tradeoff between the two error metrics, showing a characteristic L-shaped profile that indicates system performance. A gray dashed diagonal line represents the Equal Error Rate reference line where False Acceptance Rate equals False Rejection Rate. A red circular marker indicates the Equal Error Rate point. The system achieves an Equal Error Rate of approximately 0.15%, indicating balanced false acceptance and rejection behavior. The sharp corner of the Detection Error Tradeoff curve near the origin indicates

that the system simultaneously maintains low false acceptance and false rejection rates across operating thresholds.

[0124] FIG. 14 illustrates False Acceptance Rate versus False Rejection Rate analysis according to various embodiments.

5 **[0125]** The plot in FIG. 14 displays two curves against a threshold axis ranging from 0.0 to 1.0 on the horizontal axis and an Error Rate axis ranging from 0.0 to 1.0 on the vertical axis. The False Acceptance Rate curve, shown in blue, begins near zero at low threshold values and remains relatively flat until approximately threshold 0.8, after which the False Acceptance Rate curve rises sharply toward 1.0
10 as the threshold approaches 1.0. The False Rejection Rate curve, depicted in red, exhibits the opposite behavior, starting at approximately 1.0 at low threshold values and dropping rapidly to near zero as the threshold increases beyond approximately 0.1.

[0126] A vertical dashed green line marks the intersection point of the two
15 curves. The system identifies an optimal decision threshold where False Acceptance Rate and False Rejection Rate intersect at a threshold value of 0.617. At this intersection point, both error rates are equal, representing the Equal Error Rate of 0.15 percent. The identified threshold of 0.617 provides a practical operating point for deployment where false acceptance and false rejection errors are
20 minimized and equalized.

[0127] FIG. 15 illustrates a Precision-Recall curve according to various embodiments.

[0128] The Precision-Recall curve in FIG. 15 plots Precision on the vertical
25 axis ranging from 0 to 1.0 against Recall on the horizontal axis also ranging from 0 to 1.0. The curve, shown as a solid green line, demonstrates performance by maintaining precision at approximately 1.0 across nearly the entire recall range, showing a slight decrease as recall approaches 1.0. The near-rectangular shape indicates that the classifier achieves high precision while simultaneously capturing almost all positive instances.

30 **[0129]** The system achieves an Average Precision of 0.9999 on precision-recall evaluation, as displayed on the graph. A horizontal dashed blue line represents the

baseline at approximately 0.208, which corresponds to the proportion of positive samples in the dataset and serves as a reference for random classifier performance. The area between the Precision-Recall curve and the baseline, shaded in light green, visually emphasizes the improvement of the system over random classification. The precision-recall analysis demonstrates that the system reliably identifies deepfake audio samples with minimal false positives while maintaining comprehensive detection coverage of actual deepfake instances.

[0130] FIG. 16 illustrates a Receiver Operating Characteristic curve according to various embodiments.

10 **[0131]** The Receiver Operating Characteristic curve in FIG. 16 plots the True Positive Rate on the vertical axis against the False Positive Rate on the horizontal axis, with both axes ranging from 0 to 1.0. The ROC curve, shown as an orange line, demonstrates classification performance by rising sharply from the origin to nearly perfect true positive detection while maintaining a low false positive rate. A specific coordinate point is annotated at approximately 0.003 false positive rate and 1.000 true positive rate, indicating that the system achieves complete true positive detection with minimal false alarms.

15 **[0132]** The diagonal dashed blue line represents the performance of a random classifier with an AUC of 0.50, serving as a baseline reference. The legend indicates that the threshold value is 0.799. The system achieves an AUC-ROC score of 1.000, demonstrating near-ideal classification capability. A green checkbox labeled TARGET MET with a threshold of greater than or equal to 0.95 confirms that the system has exceeded the predefined performance objective. The shaded orange region beneath the ROC curve visually represents the area under the curve, emphasizing the near-ideal separation between authentic and deepfake audio classifications achieved by the multi-domain feature fusion architecture.

25 **[0133]** FIG. 17 illustrates a normalized confusion matrix for audio deepfake classification according to various embodiments.

30 **[0134]** The normalized confusion matrix in FIG. 17 is presented as a two-by-two grid with color-coded cells representing classification performance. The matrix

is organized with true labels on the vertical axis and predicted labels on the horizontal axis, both divided into Real and Fake categories.

[0135] The top-left cell shows 99.42% in white text on a dark blue background, indicating the proportion of real audio samples correctly classified as real. The top-right cell displays 0.58% in black text on a light background, representing real samples incorrectly classified as fake. The bottom-left cell shows 0.00% in black text on a light background, indicating fake samples misclassified as real. The bottom-right cell presents 100.00% in white text on a dark blue background, showing fake samples correctly identified as fake.

[0136] The system achieves 100% recall for deepfake samples while maintaining only 0.58% misclassification rate for authentic audio. A vertical color scale bar on the right side of the matrix ranges from 0.0 at the bottom in white to 1.0 at the top in dark blue, with intermediate shades of blue representing proportional values. Below the matrix, text states Total Samples: 432, Correct: 430, Accuracy: 99.5%, providing quantitative performance metrics. The visualization uses color intensity to emphasize the strong diagonal performance, with the dark blue cells along the diagonal indicating high correct classification rates and the near-white off-diagonal cells showing minimal misclassification.

[0137] There are several technical advantages of the audio deepfake detection system of the present disclosure. First, the integration of multi-domain feature extraction through parallel spectral, acoustic, and temporal processing branches enables comprehensive analysis of audio signals that captures complementary discriminative information unavailable from any single feature representation alone. The adaptive multi-head self-attention fusion mechanism dynamically prioritizes feature contributions based on audio characteristics, providing superior adaptability to varying audio quality and diverse attack types without requiring retraining or manual parameter adjustment. The frame-level temporal analysis capability enables detection of localized anomalies within longer audio streams, allowing identification of both fully synthetic audio and partially manipulated content that would evade systems operating only at the clip level. The domain-aware weighting mechanism, which assigns empirically determined weights of 42

percent to spectral features, 35 percent to acoustic features, and 23 percent to temporal features, optimizes the fusion process for maximum detection accuracy across diverse deepfake generation techniques including text-to-speech, voice conversion, and neural audio synthesis. The system demonstrates graceful
5 performance degradation under adverse noise conditions rather than catastrophic failure, maintaining accuracy above 90 percent for moderate noise levels and above 80 percent even under extreme noise scenarios, which is critical for deployment in real-world forensic and security applications where audio quality may be compromised. The modular architecture implemented using Python and PyTorch
10 provides reproducibility, scalability, and extensibility, enabling integration into digital forensics platforms, media verification systems, and cybersecurity applications. These technical advantages collectively contribute to a robust and generalizable solution for audio deepfake detection that achieves approximately 99.5 percent accuracy, an AUC-ROC score exceeding 0.95, and an Equal Error Rate
15 of approximately 0.15 percent, significantly outperforming single-domain baseline approaches.

[0138] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates
20 otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and
25 logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field.

Dated This 20th day of February 2026

30


(Kuldeep Singh)
Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358

CLAIMS

WE CLAIM:

1. A computer-implemented method (100) for audio deepfake detection, comprising:
 - 5 receiving (102) an audio input;
 - extracting (106) spectral features from a spectral representation of the audio input using a convolutional neural network (64108);
 - extracting (108) acoustic features from cepstral coefficients of the audio input using a recurrent neural network (64110);
 - 10 extracting (110) temporal features from a raw waveform of the audio input using a one-dimensional convolutional neural network (64112);
 - fusing (112) the spectral features, the acoustic features, and the temporal features using a self-attention module (64114) to generate a fused feature representation;
 - 15 classifying (114) the fused feature representation using a neural network classifier (64116); and
 - outputting (116) a classification indicating whether the audio input is real or deepfake along with a confidence score.
2. The method (100) of claim 1, further comprising:
 - 20 preprocessing (104) the audio input through noise handling, normalization, and segmentation prior to extracting the spectral features, the acoustic features, and the temporal features.
3. The method (100) of claim 1 or 2, wherein extracting the spectral features comprises:
 - 25 generating (206) a Mel-spectrogram (128) with 128 mel bands from the audio input; and
 - processing the Mel-spectrogram (128) via the convolutional neural network (64108).
4. The method (100) of any of claims 1 to 3, wherein extracting the acoustic
30 features comprises:

computing (208) Mel-frequency cepstral coefficients (40) with 40 coefficients along with delta and delta-delta derivatives from the audio input; and processing (210) the Mel-frequency cepstral coefficients via a bidirectional long short-term memory network (64212).

5 5. The method (100) of any of claims 1 to 4, wherein extracting the temporal features comprises:

 sampling (204) 64,000 samples (64,000) from the raw waveform; and
 processing the sampled raw waveform through the one-dimensional convolutional neural network (64112).

10 6. The method (100) of any of claims 1 to 5, wherein fusing the spectral features, the acoustic features, and the temporal features comprises:

 assigning (312) domain-aware weights (42) to the spectral features, the acoustic features, and the temporal features, wherein the spectral features are assigned a weight of 42 percent, the acoustic features are assigned a weight of 35
15 percent, and the temporal features are assigned a weight of 23 percent.

 7. The method (100) of any of claims 1 to 6, further comprising:
 segmenting (404) the audio input into temporal frames;
 generating (406) a deepfake probability score for each temporal frame;
 applying (408) a threshold (0.5) of 0.5 to frame predictions; and
20 aggregating (410) the frame scores using at least one of mean, median, max, or voting methods to compute an overall confidence score.

 8. An audio deepfake detection system (64100), comprising:
 an audio input module (64102) configured to receive an audio input;
 a multi-branch feature extraction module (64106) comprising:
25 a first branch (64108) comprising a convolutional neural network configured to process a spectral representation of the audio input to generate spectral features,

 a second branch (64110) comprising a recurrent neural network configured to process cepstral coefficients of the audio input to generate acoustic features, and

a third branch (64112) comprising a one-dimensional convolutional neural network configured to process raw waveform data of the audio input to generate temporal features;

5 a self-attention fusion module (64114) configured to fuse the spectral features, the acoustic features, and the temporal features using an attention-based mechanism to generate a fused feature representation;

a classification module (64116) configured to classify the fused feature representation as real or deepfake; and

10 an output module (64118) configured to output a classification result along with a confidence score.

9. The system (64100) of claim 8, further comprising:

a preprocessing module (64104) configured to perform noise handling, normalization, and segmentation on the audio input prior to processing by the multi-branch feature extraction module (64106).

15 10. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform the method of any of claims 1 to 7.

Dated This 20th day of February 2026

20



(Kuldeep Singh)

Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358

ABSTRACT

MULTI-DOMAIN FEATURE FUSION BASED DEEP LEARNING SYSTEM
FOR AUDIO DEEPPFAKE DETECTION

The present disclosure provides a computer-implemented method (100) for audio
5 deepfake detection that includes receiving (102) an audio input, extracting (106)
spectral features from a spectral representation of the audio input using a
convolutional neural network (64108), extracting (108) acoustic features from
cepstral coefficients of the audio input using a recurrent neural network (64110),
extracting (110) temporal features from a raw waveform of the audio input using a
10 one-dimensional convolutional neural network (64112), fusing (112) the spectral
features, the acoustic features, and the temporal features using a self-attention
module (64114) to generate a fused feature representation, classifying (114) the
fused feature representation using a neural network classifier (64116), and
outputting (116) a classification indicating whether the audio input is real or
15 deepfake along with a confidence score.

(FIG. 1)

[To be published along with figure 1]

Dated This 20th day of February 2026

20



(Kuldeep Singh)

Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358